NULA DeepSleep ESP32-S3

The NULA DeepSleep ESP32-S3 is a low-power development board with 7µA deep sleep current, ideal for battery-powered IoT applications.
The **NULA DeepSleep ESP32-S3** is a powerful development board engineered for ultra-low-power IoT applications. Built around the **ESP32-S3-WROOM-1-N8R8** module, it combines a dual-core processor with 8MB of Flash and 8MB of PSRAM, providing ample resources for complex projects. The board's standout feature is its exceptionally low deep sleep current of just 7µA, making it an ideal choice for battery-powered devices where power efficiency is crucial. With integrated **Wi-Fi** and **Bluetooth** LE connectivity, the NULA DeepSleep is the perfect center-stage for your embedded projects. Due to its ample connectivity, it excels in general use applications as well, such as display driving and wireless communicating.

It features an integrated **PCF85063A** real-time clock (RTC) for precise timekeeping, even when the main processor is in deep sleep. Power management is handled by an onboard **TP4056M Li-Ion**/**Li-Po** battery charger, allowing for battery integration and charging. The board's design ensures stable performance and provides all the necessary components for a long-lasting IoT solution right out of the box. A **USB-C to USB-C** cable is included in the package, so you can start programming immediately without any additional accessories.

Power your next project with the NULA DeepSleep ESP32-S3 if you are building remote environmental sensors, asset trackers, or low-power wireless nodes. This board provides the performance and efficiency you need with its breadboard-friendly layout that makes prototyping quick and easy, while the rich feature set supports a wide range of applications.

MCU

Microcontroller	ESP32-S3
Mcu Clock (MHz)	240 MHz
Sram (KB)	512 KB
Flash (MB)	4 MB

Connectivity

Wifi	Wi-Fi 4 (802.11n)
Bluetooth	Bluetooth Classic + BLE
Wireless Module	ESP32-S3-WROOM-1(N8R8)

Interface

Qwiic Compatible	Yes
Breadboard Compatible	Yes

Power

Battery Connector	JST 2-pin connector

Onboard Battery Charger	Yes
Supply Voltage	3.3 V or 5 V
Logic Level	3.3 V
Sleep current	16 µA

Other

Dev Environments	Arduino IDE, MicroPython, ESP-IDF

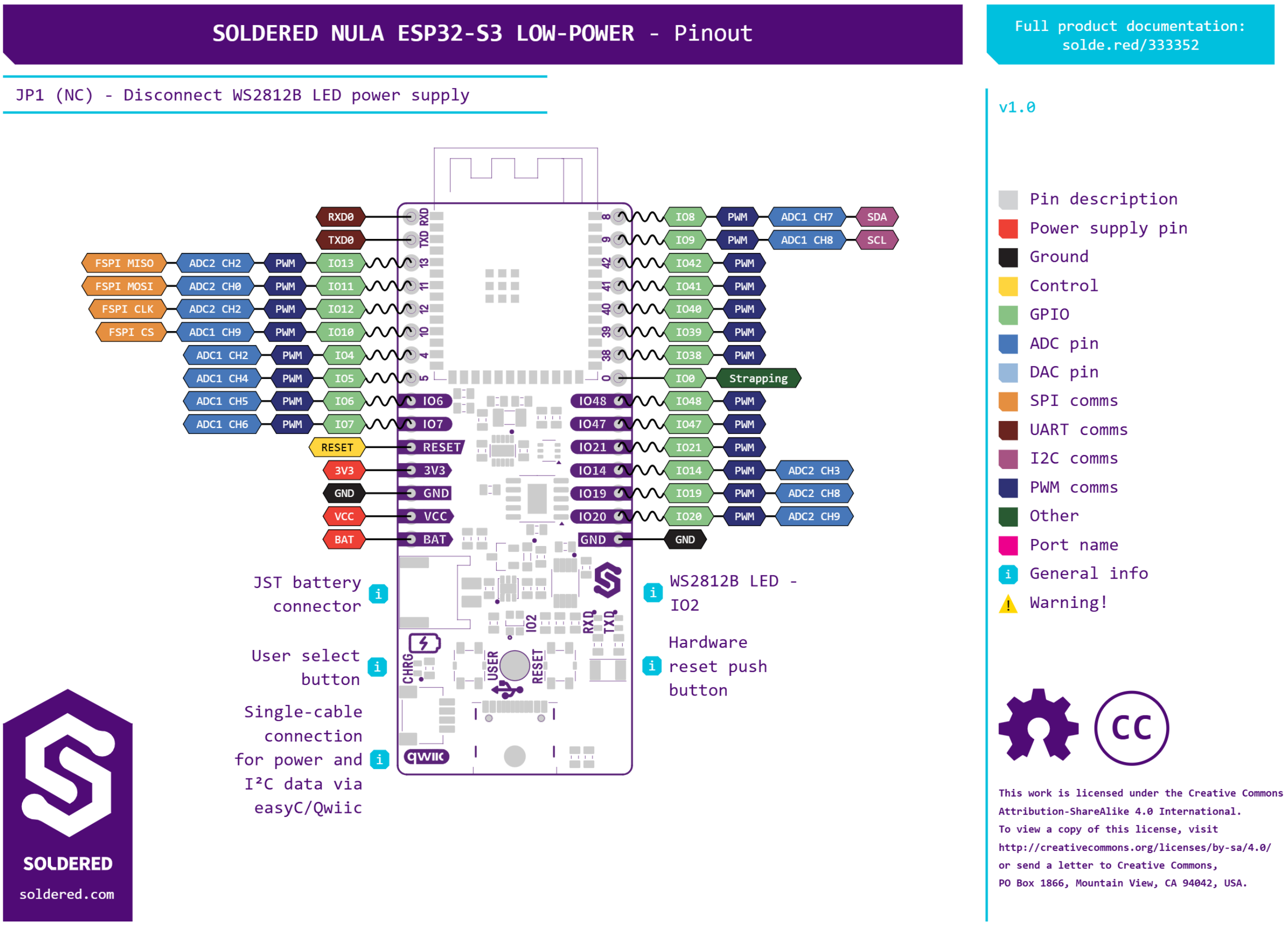
Variants

333352	No Headers
333369	Male Pin Headers

What's in the box

1× NULA DeepSleep ESP32-S3 (main product)	
1× Header pins (unsoldered)	Header pins included for all breadboard-compatible pins
1× USB-C cable	

Pinout



Typical applications

NULA DeepSleep suits battery-powered projects that spend most of their time asleep and wake briefly to take a reading or send data. Typical uses include a remote temperature logger, a mailbox or door sensor that reports over Wi-Fi, or a battery-powered prototype that needs qwiic sensors without extra wiring. The onboard battery charger lets you run it directly from a LiPo cell and recharge over USB-C. Because it supports the Arduino IDE, MicroPython, and ESP-IDF, you can start with simple example code and move to lower-level control later if your project needs longer battery life.

Resources and downloads

Hardware Details	https://docs.soldered.com/nula-deepsleep-esp32-s3/hardware-details/ (WEB)
Pinout	https://cms.soldered.com/products/333352/media/333352_pinout-png_b3cefc.png (WEB)
Full Documentation	https://docs.soldered.com/deepsleep-esp32-s3/overview/ (DOCS)